Harrison Pielke-Lombardo

Senior Backend & Full-Stack Software Engineer

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SUMMARY

Senior software engineer with 10 years of experience building production backend systems, data pipelines, and web applications in Python, Clojure, TypeScript, and Java. Most recently a core engineer on a 4-6 person team at a B2B SaaS logistics startup, building the pipeline that processed 1M+ documents per month and helping grow the product from zero to $2.1M ARR through acquisition. Owns features end to end, from design with product through on-call, and has mentored, interviewed, and served as technical contact for clients.

SKILLS

Languages: Python, Clojure, ClojureScript, TypeScript, Java, JavaScript, SQL, Bash

Backend: REST APIs, GraphQL, data pipelines, rules engines, event-driven architecture, database migrations

Frontend: TypeScript, ClojureScript, React (Reagent / re-frame), JavaScript, data visualization

Data: PostgreSQL, Datomic, DynamoDB, Redis, SQL, knowledge graphs (RDF)

Cloud & DevOps: AWS (Lambda, ECS, EC2, SQS, SNS, RDS, DynamoDB, CloudFront), Terraform, Docker, CI/CD, Git, GitHub

AI & ML: LLM integration, PyTorch, NLP, AI-assisted development (Claude, Copilot)

EXPERIENCE

Senior Software Engineer | Sep 2022 - Mar 2026

Peruse Technology Inc. (acquired), Remote

- Helped scale a B2B SaaS logistics platform from zero to $2.1M ARR and 70+ business customers as part of a 4-6 person engineering team, leading to the company's acquisition.
- Designed and built the core load processing pipeline and a custom rules engine in Clojure and Python, processing 1M+ documents per month through REST and GraphQL APIs.
- Trained and deployed LLM-based models in PyTorch to automate document processing and freight broker verification, replacing work that took customers entire back-office audit teams.
- Planned and executed the zero-downtime migration of the production database from Datomic to PostgreSQL as it approached Datomic's size limits.
- Shared ownership of event-driven AWS infrastructure as code in Terraform (Lambda, ECS, SQS, SNS, RDS); took on-call rotations and contributed to the TypeScript frontend.
- Owned features end to end, from design with the product team to release; mentored an engineer, interviewed candidates, and reviewed code.
- Served as technical contact on client calls and owned relationships with several client accounts.

Research Software Engineer (Graduate Researcher) | Aug 2016 - Aug 2022

University of Colorado Anschutz Medical Campus, Aurora, CO

- Led software for interconnected research projects across 4+ cross-functional teams, shipping 5 open-source package releases, 4 dataset releases, and 6 peer-reviewed publications.
- Built Knowtator, a Java and ClojureScript / re-frame web app for text annotation and ontology development (rewritten from a Java Swing desktop app), used to create 100k+ annotations by clinicians and researchers.
- Built data integration pipelines that merged large knowledge graphs across RDF, Datomic, and SQL stores for drug-target identification and diagnostic tools.
- Improved relation-extraction precision by 10% with a novel bootstrapping algorithm accelerated with GPU matrix operations.

OPEN SOURCE & SPEAKING

- datascript-shortest-path, a graph library with a Fibonacci heap built on immutable data structures; contributor to the re-frame ecosystem (day8/re-frame-undo).
- Invited speaker, "AI in the Transportation Sector: Challenges and Successes," Institute for Data, Econometrics, Algorithms, and Learning (IDEAL), Chicago, June 2024.

EDUCATION

M.S., Computational Bioscience, University of Colorado Anschutz Medical Campus | 2022

B.S., Applied Mathematics (minors: Computer Science, Biochemistry), University of Colorado Boulder | 2016